Math 358: Advanced Calculus Summary
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Key Results, Theorems, Definitions, etc.

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Abstract

This document contains a summary of all the key definitions, results, and theorems from class before the final exam. Point-set topology in Euclidean space; continuity and differentiability of functions in several variables. Implicit and inverse function theorems, Lagrange multipliers, Integrals in several variables, Fubini, change of variables, line and surface integrals, Green's theorem, etc...

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1. Continuity and Differentiability

1.1. Differentiability

Definition 1 (Differentiation)**.** $f : \Omega \to \mathbb{R}^m$ where $\Omega \subset \mathbb{R}^n$ is a domain[1], $f := (f_1, \ldots, f_m), f_i : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is **differentiable** at $x_0 \in \Omega$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0$$

[1] Open, Connected

Equivalently, $\forall \epsilon > 0\ \exists \delta > 0$ such that for any $x \in \mathbb{R}$ with $\|x - x_0\| < \delta$ [2] we have

$$\|f(x) - f(x_0) - L(x - x_0)\| \leq \epsilon \|x - x_0\|$$

Theorem 1. If $f : \Omega \to \mathbb{R}^m$ is differentiable at $x_0 \in \Omega$, then the linear map L is unique.

Definition 2. If f differentiable at $x_0 \in \mathbb{R}$, with $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ we write the differential map of f at $x_0 \in \Omega$ as $L = Df(x_0)$

Proposition 1. If f is differentiable at $x_0 \in \Omega$, then f is continuous at $x_0 \in \Omega$. In fact, it is Lipschitz continuous at x_0, i.e. $\exists k > 0$ and $\delta > 0$ such that if $\|x - x_0\| < \delta$, then $\|f(x) - f(x_0)\| \leq k\|x - x_0\|$

Proposition 2. If $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ differentiable at $x_0 \in \Omega$, and $f = (f_1, \ldots, f_m)$, $f_i : \Omega \to \mathbb{R}$, then f differentiable at $x_0 \in \Omega$ iff $f_j : \Omega \to \mathbb{R}$ is differentiable at $x_0 \in \Omega$ for all $j = 1, \ldots, m$.

Definition 3 (Partial Derivatives)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$, $f(x_1, \ldots, x_m) = (f_1(x_1, \ldots, x_n), \ldots, f_m(x_1, \ldots, x_n))$. Then

$$\frac{\partial f_j}{\partial x_i}(x_0) := \lim_{h \to 0} \frac{[f_j(x_1, \ldots, x_i + h, \ldots, x_n) - f_j(x_1, \ldots, x_i, \ldots, x_n)]}{h}$$

if the limit exists. We call $\frac{\partial f_j}{\partial x_i}(x_0)$ the **partial derivative** of f_j with respect to x_i at x_0.

Proposition 3. Let $f : \Omega \subset \mathbb{R}^m \to \mathbb{R}^m$, $f = (f_1, \ldots, f_m)$, $f_j : \Omega \to \mathbb{R}$, be differentiable at $x_0 \in \Omega$. Then all partial derivatives $\frac{\partial f_j}{\partial x_i}(x_0)$ exist and

$$L = Df(x_0) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \frac{\partial f_1}{\partial x_2}(x_0) & \ldots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \frac{\partial f_2}{\partial x_2}(x_0) & \ldots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \frac{\partial f_m}{\partial x_2}(x_0) & \ldots & \frac{\partial f_m}{\partial x_n}(x_0) \end{bmatrix}$$

called the Jacobian of f at x_0[3].

Remark. f differentiable at $x_0 \in \Omega \Rightarrow \frac{\partial f_j}{\partial x_i}(x_0)$ exists for all i, j. However, the converse need not be true, for example the classic example of $f : \mathbb{R}^2 \to \mathbb{R}$,

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Both partial derivatives of f at $(0, 0)$ exist and are equal to 0, but f is not differentiable at $(0, 0)$.

Steps to solve: Find candidate for $Df(0, 0)$ using the partial derivatives, then show that the limit in the definition of differentiability does not go to 0 for this candidate.

Theorem 2. Let $f = (f_1, \ldots, f_m) : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$. Suppose that all partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist and are continuous at $x_0 \in \Omega$. Then f is differentiable at x_0.

Definition 4. Suppose $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ has continuous partial derivatives $\frac{\partial f}{\partial x_i}$, $i = 1, \ldots, n$ at all points $x \in \Omega$. Then f is continuously differentiable on Ω, and we write $f \in C^1(\Omega)$.

Remark. Continuity of partial derivatives is a sufficient condition for differentiability, but it is not necessary. For example, the function $f : \mathbb{R}^2 \to \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Then $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$, however, along the line $x = y^2, x = t, y = t^2$, $\frac{\partial f}{\partial x}(t^2, t) = \frac{1}{2} \neq 0$. But, f is differentiable at $(0, 0)$ with $Df(0, 0) = 0$.

[2] Here, $\|\cdot\|$ is the standard Euclidean norm on $\mathbb{R}^m$ and $\mathbb{R}^n$.

[3] also called the derivative of f at $x_0 \in \Omega$

Proposition 4 (Basic Properties of Differentiation)**.**

(1) $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ both differentiable at $x_0 \in \Omega$, then so is $F = f + g$. Moreover, $DF(x_0) = Df(x_0) + Dg(x_0)$.

(2) $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ both differentiable at $x_0 \in \Omega$, then so is $F = f \cdot g : \Omega \to \mathbb{R}$. Moreover, $DF(x_0) = f(x_0) \cdot Dg(x_0) + g(x_0) \cdot Df(x_0)$.

(3) $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ both differentiable at $x_0 \in \Omega$, and $g(x_0) \neq 0$, then so is $F = \frac{f}{g} : \Omega \to \mathbb{R}$. Moreover,
$$DF(x_0) = \frac{g(x_0) \cdot Df(x_0) - f(x_0) \cdot Dg(x_0)}{g(x_0)^2}$$

(4) $f : \Omega \subset \mathbb{R}^n \to \tilde{\Omega} \subset \mathbb{R}^m$, $g : \tilde{\Omega} \subset \mathbb{R}^m \to \mathbb{R}^k$ with f differentiable at $x_0 \in \Omega$, and g differentiable at $y_0 = f(x_0) \in \tilde{\Omega}$. Then $H = g \circ f$ is differentiable at $x_0 \in \Omega$ and
$$DH(x_0) = Dg(y_0) \cdot Df(x_0)$$

Theorem 3 (Chain Rule)**.** Let $f : \Omega \subset \mathbb{R}^n \to \tilde{\Omega} \subset \mathbb{R}^m$, $\tilde{\Omega} = f(\Omega)$, $g : \tilde{\Omega} \subset \mathbb{R}^m \to R^k$. If f is differentiable at $x_0 \in \Omega$ and g is differentiable at $y_0 = f(x_0) \in \tilde{\Omega}$, then $H = g \circ f : \Omega \subset \mathbb{R}^n \to R^k$ is differentiable at $x_0 \in \Omega$, and
$$DH(x_0) = Dg(y_0) \cdot Df(x_0)$$

Example 1.1. Given f differentiable in R^2 and $g(r, \theta) = (r\cos\theta, r\sin\theta)$, with $(r, \theta) \in (0, \infty) \times [0, 2\pi]$. Set $F(r, \theta) = f(g(r, \theta))$, compute $\frac{\partial F}{\partial r}(r, \theta)$ and $\frac{\partial F}{\partial \theta}(r, \theta)$ in terms of the partial derivatives of f, using the chain rule.

To solve, the multivariable chain rule gives us
$$\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r} = \frac{\partial f}{\partial x}\cos\theta + \frac{\partial f}{\partial y}\sin\theta$$
and
$$\frac{\partial F}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta} = -r\frac{\partial f}{\partial x}\sin\theta + r\frac{\partial f}{\partial y}\cos\theta$$

1.2. Tangent Planes and Directional Derivatives

Definition 5 (Gradient)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable on Ω. Then $Df(x) = (\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_n}(x)) = \nabla f(x)$ is called the **gradient** of f at $x \in \Omega$.

Definition 6 (Tangent Planes)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable on Ω. $S := \operatorname{graph} f = \{(x, z) \in \Omega \times \mathbb{R} : z = f(x)\}$. Then for $x_0 \in \Omega$, $T_{x_0}S = \{(x, z) \in \mathbb{R}^n \times \mathbb{R} : z = f(x_0) + Df(x_0) \cdot (x - x_0)\}$

Definition 7 (Directional Derivative)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable on Ω. Then for $x_0 \in \Omega$ and $v \in \mathbb{R}^n$, the directional derivative of f at x_0 in the direction of v is defined as
$$D_v f(x_0) = \langle \nabla f(x_0), v \rangle$$

Proposition 5. Suppose $\nabla f(x) \neq 0$. Then $\nabla f(x)$ points in the direction of the steepest increase of f. [4]

[4] Proof follows from the Cauchy-Schwarz inequality, $< a, b >= \|a\| \cdot \|b\| \cos\theta$

1.3. Clairaut's Theorem

Theorem 4 (MVT in $\mathbb{R}^n$)**.** Let $B \subset \mathbb{R}^n$ be a ball, $B = \{x \in \mathbb{R}^n : \|x - x_0\| < p \text{ for some } x_0\}$, and $f : B \to \mathbb{R}$ differentiable on B. Then, for any $x, y \in B$ there exists $z \in B$ such that

$$f(y) - f(x) = Df(z) \cdot (y - x)$$

As a consequence, $\|f(y) - f(x)\| \leq \|Df(z)\| \cdot \|y - x\|$.

Definition 8. We say f twice differentiable at $x \in \Omega$ if Df exists locally near $x \in \Omega$ and $Df : \Omega \subset \mathbb{R}^n \to \mathbb{R}^{n \times m}$ is differentiable at $x \in \Omega$. Then $D^2 f = D(Df)$ at $x \in \Omega$. Similarly, for all $k \geq 2$, $D^k f = D(D^{k-1} f)$.

Definition 9. Given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, we say that $f \in C^k(\Omega)$ if all the partial derivatives up to order k are continuous on Ω.

Definition 10. (Hessian) Given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, the **Hessian** matrix of f at $x \in \Omega$ is

$$Hf(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(x) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(x) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(x) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(x) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(x) & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(x) & \frac{\partial^2 f}{\partial x_n \partial x_2}(x) & \dots & \frac{\partial^2 f}{\partial x_n \partial x_n}(x) \end{bmatrix}$$

For $f(x, y)$, $Df = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ and

$$D^2 f = Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x \partial x} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y \partial y} \end{bmatrix}$$

Theorem 5 (Clairaut's Theorem)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at $x \in \Omega$. Then,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$$

for all $i, j = 1, \dots, n$. Moreover, if $f \in C^2(\Omega)$, then the above holds for all $x \in \Omega$.

Corollary 1. If $\frac{\partial^2 f}{\partial x_i \partial x_j}(x)$ are all continuous at $x \in \Omega$ for all $i, j = 1, \dots, n$, then $\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$ for all $i, j = 1, \dots, n$.

1.4. Inverse Function Theorem and Implicit Function Theorem

Theorem 6 (Banach Fixed Point)**.** Let (X, d) be a complete metric space, and $f : X \to X$ be a contraction mapping with $d(f(x), f(y)) \leq \alpha d(x, y)$ for some $0 < \alpha < 1$. Then $\exists$ unique $x_0 \in X$ such that $f(x_0) = x_0$.

Definition 11. Let $M_n := \{n \times n \text{ matrices}\} \cong \mathbb{R}^{n^2}$ and $\|A\| := \sqrt{\sum_{i,j} a_{ij}^2}$ for $A = (a_{ij}) \in M_n$.

Let $GL_n := \{A \in M_n : \det A \neq 0\} = \det^{-1}(\mathbb{R} \setminus \{0\})$ Thus since $\det : M_n \to \mathbb{R}$ is continuous, GL_n is an open subset of M_n.

Theorem 7 (Inverse Function)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ be C^1. Let $x_0 \in \Omega$ and assume $Df(x_0) \in GL_n$. Then there exists neighbourhoods U and V of x_0 and $f(x_0)$ respectively such that $f(U) = V$ and $f|_U$ has a C^1 inverse map $f^{-1} : V \to U$. Moreover, for any $y \in V$, and $x = f^{-1}(y)$, we have $Df^{-1}(y) = [Df(x)]^{-1}$.

Remark. If $f \in C^k$, $k \geq 1$, then the inverse map f^{-1} is also in C^k.

Example 1.2. IFT is local. For example, $f(x,y) = (e^y \cos x, e^y \sin x)$. Then $Df(x,y) = \begin{bmatrix} -e^y \sin x & e^y \cos x \\ e^y \cos x & e^y \sin x \end{bmatrix}$, and $\det Df(x,y) = -(\cos^2 x + \sin^2 x)e^{2y} = -e^{2y} \neq 0$ for all $(x,y) \in \mathbb{R}^2$. However, $f(x + 2k\pi, y) = f(x,y)$ for all $k \in \mathbb{Z}$, so f is not injective globally on (x,y).

Theorem 8 (Implicit Function Theorem)**.** $F : \Omega \subset \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ C^k map, $X = (x,y) \in R^n \times \mathbb{R}^m$ and $X_0 = (x_0, y_0)$ with $F(X_0) = 0$. Writing $F = (F_1, \dots, F_m)$, assume that

$$D_y F(X_0) = \begin{bmatrix} \frac{\partial F_1}{\partial y_1} & \frac{\partial F_1}{\partial y_2} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \frac{\partial F_2}{\partial y_1} & \frac{\partial F_2}{\partial y_2} & \cdots & \frac{\partial F_2}{\partial y_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \frac{\partial F_m}{\partial y_2} & \cdots & \frac{\partial F_m}{\partial y_m} \end{bmatrix} (X_0) \in GL_m$$

Then there exists neighbourhoods U and V of $x_0 \in \mathbb{R}^n$ and $y_0 \in \mathbb{R}^m$ and a unique C^k map $f : U \to V$ such that $F(x, f(x)) = 0$, for all $x \in U$.

Corollary 2. Let $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be $C^k(\Omega)$. Let $x = (x', y) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and suppose $(x'_0, y_0) \in \Omega$ with $\frac{\partial f}{\partial y}(x'_0, y_0) \neq 0$. Then there exists neighbourhoods U, V of $x'_0 \in \mathbb{R}^{n-1}$ and $y_0 \in \mathbb{R}$ and a unique function $f : U \to V$, $f \in C^k(U)$ such that $\{F(x', y) = 0\} = \{y = f(x')\}\, \forall x' \in U$.

Theorem 9 (Morse Lemma)**.** $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ a C^k function $k \geq 3$. Let $0 \in \Omega$ be a critical point, i.e. $\nabla f(0) = 0$, and assume $f(0) = 0, \nabla f(0) = 0, Hf(0) \in GL_n$. Then there exists open sets U and V of $0 \in U \cap V$ and $g \in C^{k-2}(U)$, $g : U \to V$ with $g^{-1} : U \to V$, $g^{-1} \in C^2(V)$ such that $f(g(y)) = y_{\ell+1}^2 + \cdots + y_n^2 - (y_1^2 + y_2^2 + \dots y_\ell^2)$. Here $\ell \in \mathbb{Z}$, $\ell \in [0, n]$.

1.5. Taylor's Theorem

Theorem 10 (Taylor's Theorem in $\mathbb{R}^n$)**.** Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $f \in C^{k+1}(\Omega)$. Then, for any $x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^{k} \frac{1}{j!} \left[\sum_{i_1=1}^{n} \cdots \sum_{i_j=1}^{n} \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x^0)(x_{i_1} - x^0_{i_1}) \dots (x_{i_j} - x^0_{i_j}) \right] + R_k(f)(x)$$

where $\frac{|R_k(f)(x)|}{\|x - x^0\|^k} \leq M_k \|x - x^0\|$ as $x \to x^0$.

Remark. The k-th order Taylor polynomial of f at x^0 is the sum of the first two terms in the above expansion, i.e.

$$P_k(f)(x) = f(x^0) + \sum_{j=1}^{k} \frac{1}{j!} \left[\sum_{i_1=1}^{n} \cdots \sum_{i_j=1}^{n} \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x^0)(x_{i_1} - x^0_{i_1}) \dots (x_{i_j} - x^0_{i_j}) \right]$$

This is a good approximation to $f(x)$ near x^0 provided $\lim_{x \to x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0$.

Theorem 11. Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be $C^{k-1}(\Omega)$ and assume $D^{k-1} f$ is differentiable at $x^0 \in \Omega$. Then the k-th order Taylor expansion in the previous theorem still holds, but with the only assumption of the remainder term being $\lim_{x \to x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0$.

Example 1.3. Let $f(x,y) = e^x + sin(xy)$. Compute the second order Taylor expansion of f around $(0,0)$.

To solve this problem, we first recognize that $f \in C^\infty(\mathbb{R}^2)$, so we can apply Taylor's theorem. We compute the first and second order partial derivatives of f at $(0,0)$:

$$f_x = \frac{\partial f}{\partial x} = e^x + y\cos(xy), \quad f_y = \frac{\partial f}{\partial y} = x\cos(xy)$$

Therefore $f_x(0,0) = 1$ and $f_y(0,0) = 0$. Next, we compute the second order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = e^x - y^2 \sin(xy), \quad f_{yy} = \frac{\partial^2 f}{\partial y^2} = -x^2 \sin(xy), \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \cos(xy) - xy \sin(xy)$$

Therefore, $f_{xx}(0,0) = 1$, $f_{yy}(0,0) = 0$, and $f_{xy}(0,0) = 1$. Thus, the second order Taylor expansion of f around $(0,0)$ is

$$f(x,y) = f(0,0) + f_x(0,0)x + f_y(0,0)y + \frac{1}{2!}\left[f_{xx}(0,0)x^2 + 2f_{xy}(0,0)xy + f_{yy}(0,0)y^2\right] + R_2(f)(x,y)$$

which simplifies to

$$f(x,y) = 1 + x + \frac{x^2}{2} + xy + R_2(f)(x,y)$$

and the remainder term satisfies $\frac{|R_2(f)(x,y)|}{x^2+y^2} \leq M\sqrt{x^2+y^2}$ as $(x,y) \to (0,0)$.

1.6. Lagrange Multipliers

Theorem 12 (Lagrange Multiplier)**.** Let $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be C^1 functions in $\mathbb{R}$. Let $\Sigma = \{x \in \Omega : g(x) = 0\}$ and assume that $Dg(x_0) \neq 0$ for all $x_0 \in \Sigma$. Then if $f|_\Sigma$ has a max or min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda g(x_0)$$

Theorem 13 (Max/Min)**.** Let $\tilde{\Omega} \supset \Omega$ be a slightly larger domain and assume $f, g \in C^1(\Omega)$, where $\partial\Omega = \left\{x \in \tilde{\Omega} : g(x) = 0\right\}$, $\nabla g(x) \neq 0 \forall x \in \partial\Omega$. Then,

(1) If $x_0 \in \partial\Omega$ is a local max/min of $f|_{\partial\Omega}$, then $\nabla f(x_0) = \lambda \nabla g(x_0)$ for some $\lambda \in \mathbb{R}$.

(2) If $x_0 \in \mathrm{Int}(\Omega)$ is a local max/min of f in $\mathrm{Int}(\Omega)$, then $\nabla f(x_0) = 0$.

(3) Global max/min of $f|_{\tilde{\Omega}}$ is largest/smallest value of f at all points in 1 and 2.

Theorem 14 (Multiple Constraints)**.** $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $g_i : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, k$ are C^1 functions. Let $\Gamma_0 = \{x \in \Omega : g_i(x) = 0, i = 1, \ldots, k\}$ and assume that $\{\nabla g_1(x), \ldots, \nabla g_k(x)\}$ are linearly independent for all $x \in \Gamma_0$. Then if f_{Γ_0} has a max/min at $x_0 \in \Gamma_0$, then there exists $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$ such that

$$\nabla f(x_0) = \sum_{i=1}^{k} \lambda_i \nabla g_i(x_0)$$

Example 1.4. Find the absolute max and min of $f(x_1, \ldots, x_n) = x_1 + x_2$ on $B_1(0) = \left\{|x|^2 \leq 1\right\} = \left\{x_1^2 + \cdots + x_n^2 \leq 1\right\}$, $\partial B_1(0) = \left\{x_1^2 + \cdots + x_n^2 = 1\right\}$.

To solve, first observe $f \in C^k(\mathbb{R}^n)$ for all $k \geq 1$. Thus we will apply the max/min theorem with $g(x) = |x|^2 - 1 = x_1^2 + \cdots + x_n^2 - 1$. Then, since $\nabla f(x) = (1, 1, 0, \ldots, 0) \neq 0$, there are no critical points in $\mathrm{Int}(B_1(0)) = \left\{|x|^2 < 1\right\}$. So the global max/min must occur on $\partial B_1(0)$.

By Lagrange multiplier theorem, at max/min $x_0 \in \partial B_1(0)$, there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0) \Leftrightarrow (1, 1, 0, \ldots, 0) = \lambda(2x_1^0, 2x_2^0, 2x_3^0, \ldots, 2x_n^0) \tag{1}$$

$$\Leftrightarrow x_1^0 = x_2^0 = \frac{1}{2\lambda}, x_3^0 = \cdots = x_n^0 = 0 \tag{2}$$

Also,

$$(x_1^0)^2 + (x_2^0)^2 = 1 \Leftrightarrow \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 1 \Leftrightarrow \lambda = \pm\frac{1}{\sqrt{2}}$$

Thus, the candidates for max/min are $x_0 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \ldots, 0\right)$ and $x_0 = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \ldots, 0\right)$. Evaluating f at these points, we get

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \ldots, 0\right) = \sqrt{2}, \quad f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \ldots, 0\right) = -\sqrt{2}$$

Thus, the absolute max of f on $B_1(0)$ is $\sqrt{2}$ and the absolute min is $-\sqrt{2}$.

2. Riemann Integration

2.1. Integration over Box domains

Definition 12 (Partition)**.** Let $B = [a_1, b_1] \times \cdots \times [a_n, b_n]$. A **partition** of B is an ordered set

$$\mathcal{P} = \left\{P_1 = \{x_1^0, x_1^1, \ldots, x_1^n\}, \ldots, P_n = \{x_n^0, x_n^1, \ldots, x_n^m\}\right\} \quad [5]$$

Here, for any given j, P_j is the set of x_j^i parition of $[a_j, b_j]$ s.t.

$$a_j = x_j^0 < x_j^1 < \cdots < x_j^{m_j - 1} < x_j^{m_j} = b_j$$

Definition 13 (Refinement)**.** Given two partitions $\mathcal{P}$ and $\tilde{\mathcal{P}}$, we say that $\tilde{\mathcal{P}}$ is a **refinement** of $\mathcal{P}$, i.e. $\tilde{\mathcal{P}} \supset \mathcal{P}$, if for each $j = 1, \ldots, n$,

$$P_j = \left\{x_j^0, x_j^1, \ldots, x_j^{m_j}\right\} \subset \left\{\tilde{x}_j^0, \tilde{x}_j^1, \ldots, \tilde{x}_j^{m_j}\right\} = \tilde{P}_j$$

Definition 14 (Sub-box)**.** Given partition $\mathcal{P}$ of B, $\mathcal{P} = \{P_1, \ldots, P_j\}$, let

$$B_{i_1, i_2, \ldots, i_n} = [x_1^{i_1 - 1}, x_1^{i_1}] \times \cdots \times [x_n^{i_n - 1}, x_n^{i_n}]$$

be a **sub-box** of B for some $\{i_1, \ldots, i_n\}$ partition indices, i.e. $1 \leq i_j \leq m_j$ for all $j = 1, \ldots, n$. Then define the length of the jth side of the sub-box as

$$\Delta x_j^{i_j} := x_j^{i_j} - x_j^{i_j - 1} \quad \forall j$$

Thus the volume is intuitively defined as

$$\Delta V_{i_1, \ldots, i_n} := \Delta x_1^{i_1} \cdot \cdots \cdot \Delta x_n^{i_n} = \prod_{j=1}^{n} (x_j^{i_j} - x_j^{i_j - 1})$$

Definition 15 (Upper and Lower Riemann Sums)**.** Let $f : B \to \mathbb{R}$ be bounded and $\mathcal{P}$ a partition of B. Define the **lower** and **upper** Riemann sums of f with respect to $\mathcal{P}$ as

$$L(f, \mathcal{P}) = \sum_{i_1=1}^{m_1} \cdots \sum_{i_n=1}^{m_n} \left(\inf_{x \in B_{i_1, \ldots, i_n}} f(x)\right) \cdot \Delta V_{i_1, \ldots, i_n} \quad \text{and} \quad U(f, \mathcal{P}) = \sum_{i_1=1}^{m_1} \cdots \sum_{i_n=1}^{m_n} \left(\sup_{x \in B_{i_1, \ldots, i_n}} f(x)\right) \cdot \Delta V_{i_1, \ldots, i_n}$$

Lemma 15. Let $f : B \to \mathbb{R}$ be bounded, and $\tilde{\mathcal{P}} \supset \mathcal{P}$ be a refinement. Then

$$L(f, \mathcal{P}) \leq L(f, \tilde{\mathcal{P}}) \leq U(f, \tilde{\mathcal{P}}) \leq U(f, \mathcal{P})$$

Definition 16. Let $f : B \to \mathbb{R}$ be bounded, i.e. $\exists m, M$ s.t. $m \leq f(x) \leq M$, $\forall x \in \mathbb{R}$. Then define

$$U(f) := \inf_{\mathcal{P}} U(f, \mathcal{P}) \quad \text{and} \quad L(f) := \sup_{\mathcal{P}} L(f, \mathcal{P})$$

[5] The lower index denotes the dimension, and the upper index denotes the parition index.

Definition 17. Let $f : B \to \mathbb{R}$ be bounded. Then we say f **Riemann integrable** over B provided $L(f) = U(f)$, and we define

$$\int_B f(x)\, dV := L(f) = U(f)$$

Lemma 16. Let $f : B \to \mathbb{R}$ be bounded. Then f is Riemann integrable over B iff $\forall \epsilon > 0$, $\exists \mathcal{P}$ partition of B such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \epsilon$.

Theorem 17. Let $f : B \to \mathbb{R}$ be $C(B)$. Then f is Riemann integrable over B. (Continuity implies integrability)

Lemma 18 (MVT for Integrals)**.** Let $f : B \to \mathbb{R}$ be continuous. Then $\exists x_0 \in B$ s.t. $\int_B f(x)\, dV = f(x_0) \cdot \mathrm{Vol}(B)$, where $\mathrm{Vol}(B) = \int_B 1\, dV$.

2.2. Integration over General Domains

Definition 18. Given $\Omega \subset \mathbb{R}^n$ bounded, $f : \Omega \to \mathbb{R}$ bounded, we let $B \subset \mathbb{R}^n$ be a box. We define $F : B \to \mathbb{R}$ as

$$F(x) := \begin{cases} f(x) & x \in \Omega \\ 0 & x \in B \setminus \Omega \end{cases}$$

Then

$$\int_\Omega f\, dV = \int_B F\, dV \quad \text{when the latter exists}$$

Definition 19 (Content Zero)**.** $\Gamma \subset \mathbb{R}^n$ is of **Content 0** if $\forall \epsilon > 0$, there exists open box domains $B_1^0, \ldots, B_N^0$ s.t.

(1) $\Gamma \subset \bigcup_{i=1}^N B_i^0$

(2) $\sum_{i=1}^N \mathrm{Vol}(B_i^0) < \epsilon$

Example 2.1. If $\Gamma_1, \ldots, \Gamma_n$ are all content 0, then $\mathrm{Content}(\Gamma_1 \cup \cdots \cup \Gamma_N) = 0$

To solve, let $\epsilon > 0$. Since Γ_i is of content 0 for all $i = 1, \ldots, N$, there exists open box domains $B_1^i, \ldots, B_{M_i}^i$ s.t. $\Gamma_i \subset \bigcup_{j=1}^{M_i} B_j^i$ and $\sum_{j=1}^{M_i} \mathrm{Vol}(B_j^i) < \frac{\epsilon}{N}$ for all i. Then we have

$$\Gamma_1 \cup \cdots \cup \Gamma_N \subset \bigcup_{i=1}^N \bigcup_{j=1}^{M_i} B_j^i$$

and

$$\sum_{i=1}^N \sum_{j=1}^{M_i} \mathrm{Vol}(B_j^i) < \sum_{i=1}^N \frac{\epsilon}{N} = \epsilon$$

Theorem 19. Suppose $f : B \subset \mathbb{R}^n \to \mathbb{R}$ bounded. Let $\Gamma(f) =$ discontinuity set of f.

Then $\mathrm{Content}(\Gamma(f)) = 0$ iff f is Riemann integrable over B.

Example 2.2. $\Omega \subset \mathbb{R}^{n+1}$ compact. $g : \Omega \to \mathbb{R}$ continuous. Then the graph of g, $\Gamma = \{(x', g(x')) : x' \in \Omega\}$, satisfies $\mathrm{Content}(\Gamma) = 0$.

To solve,

Corollary 3. Suppose $f : B \to \mathbb{R}$ is bounded and $\Gamma(f)$ is the union of finitely many graphs of continuous functions over closed domains in $\mathbb{R}^{n-1}$. Then f is Riemann integrable over B.

Corollary 4. $\Omega \subset \mathbb{R}^n$ closed and bounded, $\partial\Omega$ is the union of finitely many graphs of C^0 functions over closed domains in $\mathbb{R}^{n-1}$. Suppose $f : \Omega \to \mathbb{R}$ bounded and $\mathrm{Content}(\Gamma(f) \cap \Omega) = 0$. Then f is Riemann integrable over Ω.

Corollary 5. Suppose Ω closed and bounded, $\Omega \subset \tilde{\Omega}$ with $G \in C^1(\tilde{\Omega})$ s.t. $\Omega = \{G(x) \leq 0 : x \in \tilde{\Omega}\}$ and $\partial\Omega = \{G(x) = 0\}$ with $\nabla G(x) \neq 0$ for all $x \in \partial\Omega$. Suppose $f : \Omega \to \mathbb{R}$ is bounded and $\mathrm{Content}(\Gamma(f) \cap \Omega) = 0$. Then f is Riemann integrable over Ω.

2.3. Basic Properties of Integrals

Proposition 6. Assume $\Omega \subset \mathbb{R}^n$ bounded and $f, g : \Omega \to \mathbb{R}$ are both Riemann integrable over Ω. Then,

(1) $\int_\Omega (f+g)\, dV = \int_\Omega f\, dV + \int_\Omega g\, dV$;

(2) $\int_\Omega \alpha f\, dV = \alpha \int_\Omega f\, dV$ for any $\alpha \in \mathbb{R}$;

(3) $f \leq g \quad \Rightarrow \quad \int_\Omega f\, dV \leq \int_\Omega g\, dV$;

(4) If $\Omega = \Omega_1 \cup \Omega_2$, Ω bounded and $\text{Content}(\Omega_1 \cap \Omega_2) = 0$. Then $\int_\Omega f\, dV = \int_{\Omega_1} f\, dV + \int_{\Omega_2} f\, dV$.

2.4. Fubini's Theorem

Definition 20. Given $f : B \to \mathbb{R}$ bounded, we define the iterated integrals as

$$\int_a^b \int_c^d f(x,y)\, dy\, dx = \int_a^b \left(\int_c^d f(x,y)\, dy \right) dx \quad \text{and} \quad \int_c^d \int_a^b f(x,y)\, dx\, dy = \int_c^d \left(\int_a^b f(x,y)\, dx \right) dy$$

Theorem 20 (Fubini for box domains)**.** Suppose $f : B \to \mathbb{R}$ is Riemann integrable over B, where $B = [a,b] \times [c,d]$.

(1) Assume $\int_c^d f(x,y)\, dy$ exists $\forall x \in [a,b]$. Then $\int_a^b \int_c^d f(x,y)\, dy\, dx$ exists and equals $\int_B f(x,y)\, dV$.

(2) Assume $\int_a^b f(x,y)\, dx$ exists $\forall y \in [c,d]$. Then $\int_c^d \int_a^b f(x,y)\, dx\, dy$ exists and equals $\int_B f(x,y)\, dV$.

2.5. Fubini for General Domains

Definition 21 (y-simple, x-simple domains)**.** A domain $\Omega \subset \mathbb{R}^2$ is called **y-simple** if it is of the form

$$\Omega = \{a \leq x \leq b : \varphi_1(x) \leq y \leq \varphi_2(x)\}$$

where $\varphi_1, \varphi_2 \in C^0([a,b])$.

Similarly, $\Omega \subset \mathbb{R}^2$ is called **x-simple** if it is of the form

$$\Omega = \{c \leq y \leq d : \psi_1(y) \leq x \leq \psi_2(y)\}$$

where $\psi_1, \psi_2 \in C^0([c,d])$. $\Omega \subset \mathbb{R}^2$ is called **simple** if it is both y-simple and x-simple.

Theorem 21 (Fubini)**.** Suppose $\Omega \subset \mathbb{R}^2$ bounded and f integrable over Ω. Then,

(1) If Ω is y-simple and $\forall x \in [a,b]$, $\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\, dy$ exists, then

$$\int_\Omega f(x,y)\, dV = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\, dy\, dx$$

(2) If Ω is x-simple and $\forall y \in [c,d]$, $\int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\, dx$ exists, then

$$\int_\Omega f(x,y)\, dV = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\, dx\, dy$$

If Ω is simple, then both hold.

Definition 22. $\Omega \subset \mathbb{R}^2$ is **elementary** if Ω is x-simple or y-simple.

Example 2.3. Compute $\iint_\Omega x\ dA$ where $\Omega = \{(x,y) \in \mathbb{R}^2 : 0 \le x \le \sqrt{\pi}, 0 \le y \le \sin x^2\}$

To solve, first note that $f(x,y) = x$ is C^0 on Ω. Since Ω is bounded, $\iint_\Omega x\ dV$ exists. Also, Ω is y-simple with $\varphi_1(x) = 0, \varphi_2(x) = \sin x^2$, $\varphi_1, \varphi_2 \in C^0([0, \sqrt{\pi}])$. By Fubini,

$$\iint_\Omega x\ dA = \int_0^{\sqrt{\pi}} \int_0^{\sin x^2} x\ dy\ dx = \int_0^{\sqrt{\pi}} x \sin x^2\ dx = \frac{1}{2}\int_0^{\pi} \sin u\ du = \frac{1}{2}\left[-\cos u\right]_0^{\pi} = 1$$

Example 2.4. Compute $I = \int_0^1 \int_y^1 e^{x^2}\ dx\ dy$.

To solve, first note $\int_y^1 e^{x^2}\ dx$ doesn't exist in closed-form. But since $e^{x^2} \in C^0(\Omega)$, $\iint_\Omega e^{x^2}\ dA$ exists. Since Ω is simple, by Fubini,

$$\iint_\Omega e^{x^2}\ dA = \int_0^1 \int_0^x e^{x^2}\ dy\ dx = \int_0^1 xe^{x^2}\ dx = \frac{1}{2}\int_0^1 e^u\ du = \frac{1}{2}(e-1)$$

Definition 23. Some notation: Given a box domain $B \subset \mathbb{R}^n$, write

$$\underline{\int_B} f\ dV := L(f) = \sup_{\mathcal{P}} L(f, \mathcal{P}) \quad \text{and} \quad \overline{\int_B} f\ dV := U(f) = \inf_{\mathcal{P}} U(f, \mathcal{P})$$

Theorem 22. Let $B = B_1 \times B_2$ box domain where $B_1 \subset \mathbb{R}^k$ and $B_2 \subset \mathbb{R}^{n-k}$ also a box. Suppose $f : B \to \mathbb{R}$ is Riemann integrable. Set

$$\ell(x) = \underline{\int_{B_2}} f(x,y)\ dV(y) \quad u(x) = \overline{\int_{B_2}} f(x,y)\ dV(y) \quad \forall x \in B_1 \subset \mathbb{R}^k$$

and

$$\tilde{\ell}(y) = \underline{\int_{B_1}} f(x,y)\ dV(x) \quad \tilde{u}(y) = \overline{\int_{B_1}} f(x,y)\ dV(x) \quad \forall y \in B_2 \subset \mathbb{R}^{n-k}$$

Then $\ell(x), u(x)$ are Riemann integrable over B_1. Also $\tilde{\ell}(y), \tilde{u}(y)$ are Riemann integrable over B_2. Moreover,

$$\int_B f(x,y)\ dV = \int_{B_1} \ell(x)\ dV(x) = \int_{B_1} u(x)\ dV(x)$$
$$\int_B f(x,y)\ dV = \int_{B_2} \tilde{\ell}(y)\ dV(y) = \int_{B_2} \tilde{u}(y)\ dV(y)$$

Definition 24 (z-simple)**.** A bounded domain $W \subset \mathbb{R}^3$ is called **z-simple** if $\exists$ an elementary region $\Omega \subset \mathbb{R}^2$ and $\eta_1, \eta_2 \in C^0(\Omega)$ s.t.

$$W = \{(x,y,z) \in \mathbb{R}^3 : (x,y) \in \Omega \text{ and } \eta_1(x,y) \le z \le \eta_2(x,y)\}$$

It is possible to permute the coordinates and define x-simple and y-simple domains in $\mathbb{R}^3$ similarly.

Theorem 23 (Fubini in $\mathbb{R}^3$)**.** Suppose $W \subset \mathbb{R}^3$ is bounded, z-simple, and $f : W \to \mathbb{R}$ os Riemann integrable over W. Assume $\forall (x,y) \in \Omega$,

$$\int_{\eta_1(x,y)}^{\eta_2(x,y)} f(x,y,z)\ dz \quad \text{exists}$$

Then it is Riemann integrable over Ω, and

$$\iiint f\ dV = \int_\Omega \left(\int_{\eta_1(x,y)}^{\eta_2(x,y)} f(x,y,z)\ dz \right) dV(x,y)$$

The same holds for x-simple and y-simple.

2.6. Change of Variables

Definition 25 (Regular Partition)**.** Let $\Omega \subset \mathbb{R}^n$ be a domain with $\text{Content}(\partial\Omega) = 0$. A **regular partition**, $\mathcal{P}_{reg} = \{D_k\}_{k=1}^N$ is a collection of subdomains $D_k \subset \Omega$; $k = 1, \dots, N$ s.t.

(1) $\Omega = \bigcup_{k=1}^N D_k$, $\text{Content}(\partial D_k) = 0 \quad \forall k = 1, \dots, N$.

(2) $\text{Content}(D_k \cap D_\ell) = 0 \quad \forall k \neq \ell$.

Definition 26. Suppose $f : \Omega \to \mathbb{R}$ is bounded, where $\Omega \subset \mathbb{R}^n$ is a bounded domain, and $\mathcal{P}_{reg} = \{D_k\}_{k=1}^N$. Define

$$U(f, \mathcal{P}_{reg}) = \sum_{k=1}^N (\sup_{D_k} f) \cdot \text{Vol}(D_k) \quad L(f, \mathcal{P}_{reg}) = \sum_{k=1}^N (\inf_{D_k} f) \cdot \text{Vol}(D_k)$$

Lemma 24. $f : B \subset \mathbb{R}^n \to \mathbb{R}$ is Riemann integrable over box B iff $\forall \epsilon > 0$, $\exists \mathcal{P}_{reg}$ regular partition of B s.t.

$$U(f, \mathcal{P}_{reg}) - L(f, \mathcal{P}_{reg}) \leq \epsilon$$

Theorem 25 (Change of Variables)**.** Let $\Omega, \Omega' \subset \mathbb{R}^n$ be bounded domains. Let $T : \Omega' \to \Omega$ be a C^1 map that is bijective from Ω' onto Ω. Then, for any $f : \Omega \to \mathbb{R}$ Riemann integrable over Ω,

$$\int_\Omega f \, dV = \int_{\Omega'} (f \circ T) \cdot |\det DT| \, dV$$

Example 2.5 (Polar Coordinates)**.** Let $x = r\cos\theta$ and $y = r\sin\theta$. $\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq a^2\}$, $\Omega' = \{(r, \theta) \in (0, a] \times [0, 2\pi)\}$ Then $T : \Omega' \to \Omega$ defined by $T(r, \theta) = (r\cos\theta, r\sin\theta)$ is a C^1 bijection. Also, $DT = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$ and $\det DT = r$. Thus, for any $f : \Omega \to \mathbb{R}$ Riemann integrable over Ω,

$$\iint_{\{x^2+y^2 \leq a^2\}} f(x, y) \, dA = \int_0^a \int_0^{2\pi} f(r\cos\theta, r\sin\theta) \cdot r \, d\theta \, dr$$

Remark. Change of variables still holds if we only assume $T : \Omega' \setminus \Gamma' \to \Omega$ is a C^1 bijection, where $\text{Content}(\Gamma') = 0$

Example 2.6. Compute $\iint_\Omega (y - x)\cos xy \, dA$ where Ω is the region defined by $x + y = 1, x + y = 4$, $xy = -1, xy = -4$, $y \geq x$.

To solve, first note that $f(x, y) = (y - x)\cos xy$ is C^0 on Ω, thus it is Riemann integrable over Ω. Now to solve, we will perform a change of variables with $u = x + y$ and $v = xy$. Then $T : \Omega' \to \Omega$ defined by $T(u, v) = (x, y)$ is a C^1 bijection, where $\Omega' = \{(u, v) : 1 \leq u \leq 4, -4 \leq v \leq -1\}$. Also, $|\det(DT^{-1})||\det \begin{bmatrix} 1 & 1 \\ y & x \end{bmatrix}| = |x - y| = \sqrt{u^2 - 4v} = \frac{1}{|\det DT|}$. Thus by change of variables,

$$\begin{aligned} \iint_\Omega (y - x)\cos xy \, dA &= \int_1^4 \int_{-4}^{-1} \sqrt{u^2 - 4v}\cos v \cdot \frac{1}{\sqrt{u^2 - 4v}} \, dv \, du \\ &= \int_1^4 \int_{-4}^{-1} \cos v \, dv \, du = \int_1^4 [\sin v]_{-4}^{-1} \, du = 3(\sin(-1) - \sin(-4)) \end{aligned}$$

Example 2.7. Compute $\iint_D e^{-(x^2+y^2)} \, dA$ where $D = \{x^2 + y^2 \leq R^2\}$.

To solve, first note that $f(x, y) = e^{-(x^2+y^2)}$ is C^0 on Ω, thus it is Riemann integrable over Ω. Now we perform a change of variables with $x = r\cos\theta$ and $y = r\sin\theta$. Then $T : \Omega' \to \Omega$ is defined as $T(r, \theta) = (r\cos\theta, r\sin\theta)$. Then T is a C^1 bijection, where $\Omega' = \{(r, \theta) : 0 \leq r \leq R, 0 \leq \theta \leq 2\pi\}$. Also, $DT = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$ and $\det DT = r$. Thus by change of variables,

$$\iint_D e^{-(x^2+y^2)} \, dA = \int_0^R \int_0^{2\pi} e^{-r^2} \cdot r \, d\theta \, dr = 2\pi \int_0^R re^{-r^2} \, dr = \pi(1 - e^{-R^2})$$

Example 2.8 (spherical coordinates)**.** Let $x = p\sin\varphi\cos\theta$, $y = p\sin\varphi\cos\theta$, $z = p\cos\theta$ Let $\Omega = \{(x,y,z) : x^2+y^2+z^2 \le R^2\}$ and $\Omega' = \{(p,\varphi,\theta) : 0 \le p \le a, 0 \le \varphi \le \pi, 0 \le \theta \le 2\pi\}$. Then for any $f : \Omega \to \mathbb{R}$ Riemann integrable over Ω,

$$\iiint_\Omega f\ dV = \iint_{\Omega'} f(p\sin\varphi\cos\theta, p\sin\varphi\sin\theta, p\cos\varphi)\cdot p^2\sin\varphi\ dp\ d\theta\ d\varphi$$

2.7. Improper Integrals

Definition 27. Suppose $A \subset \mathbb{R}^n$ a region, $f : A \to \mathbb{R}$, $f(x) \ge 0 \quad \forall x \in A$. Suppose $\forall r > 0$, f is bounded and integrable over $A \cap B_r(0)$. Then we define

$$\int_A f\ dV := \lim_{r\to\infty}\int_{B_r(0)\cap A} f\ dV$$

If it exists, is is the **improper integral** of f over A.

Definition 28. Consider $\{A_k\}_{k=1}^\infty$, $A_k \subset A$, $A_k \uparrow$. Then $\{A_k\}_{k=1}^\infty$ are **exhausting regions** of A if

(1) A_k compact for all k;

(2) $\text{Content}(\partial A_k) = 0$;

(3) $\forall r > 0$, $\exists k_0(r)$ s.t. $B_r(0)\cap A \subset A_k$, $\forall k \ge k_0(r)$.

For example, $\{B_k(0)\cap A\}_{k=1}^\infty$ are exhausting regions of A.

Theorem 26. Assume $f \ge 0$ bounded and integrable on $B_r(0)\cap A$, $\forall r > 0$. Then for any exhausting regions $\{A_k\}$ of A,

$$\lim_{k\to\infty}\int_{A_k} f\ dV = \lim_{r\to\infty}\int_{B_r(0)\cap A} f\ dV$$

Example 2.9. Evaluate $\int_0^\infty e^{-x^2}$.

To solve, first note that $f(x) = e^{-x^2}$ is bounded and integrable over $[0,r]$ for all $r > 0$. Thus $\int_0^\infty e^{-x^2}$ exists. Also, $\{[0,k]\}_{k=1}^\infty$ are exhausting regions of $[0,\infty)$. Thus,

$$\begin{aligned}\left(\int_0^\infty e^{-x^2}\ dx\right)^2 = \int_0^\infty e^{-x^2}\ dx\cdot\int_0^\infty e^{-y^2}\ dy &= \lim_{k\to\infty}\iint_{[0,k]^2} e^{-(x^2+y^2)}\ dA \\ &= \lim_{k\to\infty}\int_0^k\int_0^k e^{-(x^2+y^2)}\ dy\ dx = \lim_{k\to\infty}\frac{\pi}{2}\int_0^k re^{-r^2}\ dr = \frac{\pi}{4}\end{aligned}$$

Therefore $\int_0^\infty e^{-x^2}\ dx = \frac{\sqrt{\pi}}{2}$.

Definition 29. Given $f : A \to \mathbb{R}$, define $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$. iF both $\int_A f^+\ dV$ and $\int_A f^-\ dV$ exist, then f integrable over A and

$$\int_A f\ dV = \int_A f^+\ dV - \int_A f^-\ dV$$

3. Vector Fields, Curves, and Surfaces

3.1. Path Integrals/ Line Integrals

Definition 30. A mapping $\gamma : [a, b] \to \mathbb{R}^n$ is called $C^1([a, b])$ if $\gamma \in C^1(a, b)$ and

$$\lim_{t\to a^+} \frac{\gamma(t) - \gamma(a)}{t - a} = \gamma'(a^+) \quad \text{and} \quad \lim_{t\to b^-} \frac{\gamma(t) - \gamma(b)}{t - b} = \gamma'(b^-)$$

$\gamma : [a, b] \to \mathbb{R}^n$ is a **parameteric curve** in $\mathbb{R}^n$ if $\gamma \in C^1([a, b])$. Then

$$\gamma(t) = (x_1(t), x_2(t), \dots, x_n(t)) \quad \text{and} \quad \gamma'(t) = (x_1'(t), x_2'(t), \dots, x_n'(t)) \text{ is the tangent vector}$$

Definition 31. Let $\gamma : [a, b] \to \mathbb{R}^n$ be a continuous map. Then γ is piecewise C^1 if it is C^1 on each subinterval of a partition of $[a, b]$. i.e. it has finitely many "corners"/non-differentiable points.

We say γ is **regular** if $\gamma \in C^1([a, b])$ and $\gamma'(t) \neq 0$ for all $t \in [a, b]$. We say γ is **piecewise regular** if it is regular on each subinterval of a partition of $[a, b]$.

Proposition 7. Let $\gamma : [a, b] \to \mathbb{R}^n$ be regular. Then there exists a partition $a = t_0 < t_1 < \cdots < t_N = b$ s.t. on each $[t_{i-1}, t_i]$, $i = 1, \dots, N$, there is a j s.t. $t \mapsto x_{j_i}(t)$ is C^1 invertible and for $t \in [t_{i-1}, t_i]$, $\gamma(t) = (x_1(x_{j_i}(t)), \dots, x_{j_{i-1}}(x_{j_i}(t)), x_{j_i}(t), x_{j_{i+1}}(x_{j_i}(t)), \dots, x_n(x_{j_i}(t)))$. i.e. γ is a graph over x_{j_i}.

Definition 32. For each piecewise C^1 curve γ, we define the **length** of γ as

$$L(\gamma) := \sum_{i=1}^{n} \int_{a_{i-1}}^{a_i} \|\gamma'(t)\| \, dt = \sum_{i=1}^{n} \int_{a_{i-1}}^{a_i} \sqrt{(x_1'(t))^2 + \cdots + (x_n'(t))^2} \, dt \quad [6]$$

Definition 33 (Reparameterization)**.** Let $\gamma : [a, b] \to \mathbb{R}^n$ be C^1. $\tilde{\gamma} : [c, d] \to \mathbb{R}^n$ is a **reparameterization** of γ if there exists a C^1 bijection $h : [c, d] \to [a, b]$ s.t. $\tilde{\gamma}(t) = \gamma(h(t))$.

Note then $\tilde{\gamma}'(t) = h'(t) \cdot \gamma'(h(t))$. Thus, if γ is regular and $h'(t) > 0$ for all t, then $\tilde{\gamma}$ is also regular.

Theorem 27. Suppose $\gamma : [a, b] \to \mathbb{R}^n$ is a C^1 curve, and $\tilde{\gamma} : [c, d] \to \mathbb{R}^n$ is a C^1 reparameterization. Then $L(\gamma) = L(\tilde{\gamma})$.

Definition 34 (arc-length parameterization)**.** Let $\gamma : [a, b] \to \mathbb{R}^n$ be a C^1 regular parameterized curve. Then define

$$s(t) = \int_a^t \|\gamma'(u)\| \, du \in C^1([a, b]) \quad \text{and} \quad s'(t) = \|\gamma'(t)\| > 0 \quad \forall t \in [a, b]$$

Then $s : [a, b] \to [0, L]$ is a C^1 monotone increasing bijection where $L = L(\gamma) = \int_a^b \|\gamma'(t)\| \, dt$. Thus, s has a C^1 inverse $t(s) : [0, L] \to [a, b]$. Then $\tilde{\gamma}(s) = \gamma(t(s))$ is a reparameterization of γ and is called the **arc-length parameterization** of γ. Note that $\|\tilde{\gamma}'(s)\| = 1$ for all s.

Definition 35. Let f be C^0 in a neighbourhood of γ. The **path integral** of f along γ is defined as

$$\int_\gamma f \, ds := \int_a^b f(\gamma(t)) \cdot \|\gamma'(t)\| \, dt$$

If we instead use arc-length to parameterize the curve, then

$$\int_\gamma f \, ds = \int_0^L f(\gamma(t(s))) \, ds$$

by the change of variables formula, since $s'(t) = \|\gamma'(t)\|$

[6] Proof follows from MVT

3.2. Vector Fields and Line Integrals

Definition 36 (Vector Field)**.** A **vector field** on $\Omega \subset \mathbb{R}^n$ is a map $\vec{F} : \Omega \to \mathbb{R}^n$.

Definition 37. The **total work** done by $\vec{F}$ along γ is defined as

$$\int_\gamma \vec{F}\, ds := \lim_{N\to\infty} \sum_{i=1}^{N} \vec{F}(\gamma(t_i))\, \Delta s_i \quad \text{and} \quad = \int_a^b \vec{F}(\gamma(t)) \cdot \gamma'(t)\, dt \quad \text{assuming the integral exists.}$$

Definition 38 (Line Integral)**.** Let $\gamma \subset \mathbb{R}^n$ be a piecewise smooth path and let $\vec{F}$ be a vector field defined on a neighbourhood of γ. The **line integral** of $\vec{F}$ along γ is defined as

$$\int_\gamma \vec{F} \cdot \vec{ds} = \int_a^b \vec{F}(\gamma(t)) \cdot \gamma'(t)\, dt \quad \text{if } \gamma \text{ is regular.}$$

Note if we let $\vec{T}(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}$ be the unit tangent field, then

$$\int_\gamma \vec{F} \cdot \vec{ds} = \int_\gamma \vec{F}(\gamma(t)) \cdot \vec{T}(t)\, ds \quad \text{where } ds = \|\gamma'(t)\| dt$$

Definition 39. Let $\gamma : [a, b] \to \mathbb{R}^n$ be a piecewise C^1 path and $\tilde{\gamma} : [c, d] \to \mathbb{R}^n$ a reparameterization of γ with a bijective C^1 map $h : [c, d] \to [a, b]$ s.t. $\tilde{\gamma}(t) = \gamma(h(t))$. Then

(1) $\tilde{\gamma}$ is **orientation-preserving** if $\tilde{\gamma}(c) = \gamma(a)$ and $\tilde{\gamma}(d) = \gamma(b)$;

(2) $\tilde{\gamma}$ is **orientation-reversing** if $\tilde{\gamma}(c) = \gamma(b)$ and $\tilde{\gamma}(d) = \gamma(a)$.

Lemma 28. Let $\gamma : [a, b] \to \mathbb{R}^n$ be a piecewise smooth path and $\tilde{\gamma} : [c, d] \to \mathbb{R}^n$ a reparameterization of γ. Let $\vec{F}$ be a C^0 vector field. Then

$$\int_\gamma \vec{F} \cdot \vec{ds} = \epsilon \int_{\tilde{\gamma}} \vec{F} \cdot \vec{ds} \quad \text{where } \epsilon = 1 \text{ if } \tilde{\gamma} \text{ is orientation-preserving and } \epsilon = -1 \text{ otherwise.}$$

3.3. Conservative Vector Fields

Definition 40 (Conservative Vector Field)**.** $\vec{F} \in C^1(\Omega, \mathbb{R}^n)$ is **conservative** if $\exists f \in C^1(\Omega)$ s.t. $\vec{F} = \nabla f$.

Example 3.1. $\vec{F}(x, y) = (x, y)$, $(x, y) \in \mathbb{R}^2$ is the Radial Vector Field. Here, set $f(x, y) = \frac{1}{2}(x^2 + y^2)$. Then $\nabla f = (x, y) = \vec{F}$, so $\vec{F}$ is conservative.

Theorem 29. Suppose $\vec{F}$ is conservative in a neighbourhood of a piecewise smooth path γ. Then

$$\int_\gamma \vec{F} \cdot \vec{ds} = f(\gamma(b)) - f(\gamma(a))$$

Corollary 6. If $\vec{F}$ is conservative in a neighbourhood of a closed piecewise smooth path γ, then $\int_\gamma \vec{F} \cdot \vec{ds} = 0$.

Theorem 30. Assume $\vec{F} = (F_1, F_2, \ldots, F_n) \in C^1(\Omega, \mathbb{R}^n)$ is conservative. Then

$$\frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x) \quad \forall i, j = 1, \ldots, n, \forall x \in \Omega$$

3.4. Poincaré's Lemma

Lemma 31. Suppose $g(x,y)$ is continuous defined in $\Omega \times A$, where $\Omega \subset \mathbb{R}^n$ open and $A \subset \mathbb{R}^m$ compact. Assume in addition that $g \in C^1(\Omega \times A)$. Then

$$\frac{\partial}{\partial x_i}\left(\int_A g(x,y)\,dy\right) = \int_A \frac{\partial}{\partial x_i} g(x,y)\,dy \quad \forall i = 1,\dots,n$$

Definition 41. $\Omega \subset \mathbb{R}^n$ is **star-shaped** if $\exists p_0 \in \Omega$ s.t. the line $\ell(p_0, x)$ is contained in Ω for all $x \in \Omega$. i.e. $\forall x \in \Omega$, $\{(1-t)p_0 + tx : t \in [0,1]\} \subset \Omega$.

Theorem 32 (Poincaré's Lemma)**.** Suppose $\Omega \subset \mathbb{R}^n$ is star shaped and $\vec{F} \in C^1(\Omega, \mathbb{R}^n)$. Then $\vec{F}$ is conservative iff

$$\frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x) \quad \forall i,j = 1,\dots,n, \forall x \in \Omega$$

3.5. Surfaces and Surface Integrals

Definition 42 (Parameterized Surface)**.** A **parameterized surface** $S \subset \mathbb{R}^3$ is the image of a map $\Phi : D \to \mathbb{R}^3$,

$$S = \{(x(u,v), y(u,v), z(u,v)) = \Phi(u,v) : (u,v) \in D\}$$

Where $D \subset \mathbb{R}^2$ is some bounded domain. If Φ is C^k, we call S a C^k surface. Note that:

$$\Phi_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right) \quad \text{and} \quad \Phi_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)$$

are the tangent vectors of S at $\Phi(u,v)$. The tangent plane of S at $\Phi(u,v)$ is the plane spanned by Φ_u and Φ_v.

Definition 43 (Regular Surface)**.** We say S is a **regular surface** if $\vec{\Phi_u} \times \vec{\Phi_v} \neq \vec{0}$ for all $(u,v) \in D$.

Definition 44 (Area of a Surface)**.** Let S be a regular C^1 surface parameterized by $\Phi : D \to S$. Then the area of S is defined as

$$\text{area}(S) := \iint_D \|\vec{\Phi_u} \times \vec{\Phi_v}\|\,dA$$

Here,

$$\|\vec{\Phi_u} \times \vec{\Phi_v}\| = \sqrt{\left|\frac{\partial(x,y)}{\partial(u,v)}\right|^2 + \left|\frac{\partial(y,z)}{\partial(u,v)}\right|^2 + \left|\frac{\partial(x,z)}{\partial(u,v)}\right|^2}$$

Definition 45. Let $S = \Phi(D)$ be a C^1 parameterized surface and suppose $f \in C^0(S)$, i.e. f is C^0 in a neighbourhood of S in $\mathbb{R}^3$, i.e. $f \circ \Phi \in C^0(D, \mathbb{R}^3)$. Then the **surface integral** of f over S is defined as

$$\iint_S f\,dS := \iint_D f(\Phi(u,v)) \cdot \|\vec{\Phi_u} \times \vec{\Phi_v}\|\,du\,dv$$

Example 3.2. There is an important special case. Suppose $S = \big\{(x,y,g(x,y)) : (x,y) \in D \text{ and } g \in C^1(D)\big\}$. Here $(x,y) = (u,v)$, $\Phi(u,v) = (u,v,g(u,v))$, $\Phi \in C^1(D, \mathbb{R}^3)$. In this case

$$\vec{\Phi_x} = (1,0,g_x) \quad \text{and} \quad \vec{\Phi_y} = (0,1,g_y) \quad \text{and} \quad \|\vec{\Phi_x} \times \vec{\Phi_y}\| = \sqrt{1 + g_x^2 + g_y^2}$$

Then $dS = \sqrt{1 + g_x^2 + g_y^2}\,dx\,dy$ and so

$$\iint_S f\,dS = \iint_D f(x,y,g(x,y))\sqrt{1 + g_x^2 + g_y^2}\,dx\,dy$$

Example 3.3 (Helicoid)**.** $D = \{0 \leq \theta \leq 2\pi, 0 \leq r \leq 1\}$. $\Phi(r,\theta) = (r\cos\theta, r\sin\theta, \theta)$ is a C^1 parameterization of the helicoid. Then

$$\vec{\Phi_r} = (\cos\theta, \sin\theta, 0) \quad \text{and} \quad \vec{\Phi_\theta} = (-r\sin\theta, r\cos\theta, 1) \quad \text{and} \quad \|\vec{\Phi_r} \times \vec{\Phi_\theta}\| = \sqrt{1+r^2}$$

Definition 46. Let $S = \Phi(D)$ be a C^1 parameterized surface. We say that S is **regular** if $(\vec{\Phi_u} \times \vec{\Phi_v})(u,v) \neq \vec{0}$ for all $(u,v) \in D$. If S is regular, then we can define the **unit normal**

$$\vec{n} = \frac{\pm(\Phi_u \times \Phi_v)}{\|\Phi_u \times \Phi_v\|}$$

The choice of $\pm$ is called the **orientation** of S. Note that if S is regular, then $\vec{n}$ is well-defined and continuous on S.

Example 3.4. S is the graph of g, i.e. $S = \{(x,y,g(x,y)) : (x,y) \in D \text{ and } g \in C^1(D)\}$. Then

$$\vec{n} = \frac{\pm(-g_x, -g_y, 1)}{\sqrt{1+g_x^2+g_y^2}}$$

Definition 47 (Surface Integral of a Vector Field)**.** Let F be a C^0 vector field on an oriented regular parameterized surface. Then

$$\iint_S \vec{F}\cdot d\vec{S} := \iint_D \vec{F}(\Phi(u,v))\cdot\vec{n}\, ds = \iint_D \vec{F}(\Phi(u,v))\frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}\cdot\|\Phi_u \times \Phi_v\|\, du\, dv = \iint_D \vec{F}(\Phi(u,v))\cdot(\Phi_u \times \Phi_v)\, du\, dv$$

Remark. The definition is independent of the choice of parameterization, modulo the choice of oerientation.

Definition 48 (reparameterization of a surface)**.** Let $\Phi : D \subset \mathbb{R}^2 \to S$ be a regular C^1 parameterization, $\tilde{\Phi} : \tilde{D} \subset \mathbb{R}^2 \to S$ another regular C^1 parameterization. We say $\tilde{\Phi}$ is a **reparameterization** of Φ if there exists a C^1 bijection $h : \tilde{D} \to D$ s.t. $\tilde{\Phi}(u,v) = \Phi(h(u,v))$. Then $\tilde{\Phi}$ is **orientation-preserving** (+) if

$$\frac{\Phi_u \times \Phi_v(h(s,t))}{\|\Phi_u \times \Phi_v\| \cdot (h(s,t))} = \frac{\tilde{\Phi}_s \times \tilde{\Phi}_t(s,t)}{\|\tilde{\Phi}_s \times \tilde{\Phi}_t\|(s,t)} \quad \forall (s,t) \in \tilde{D}$$

and **orientation-reversing** (-) if

$$\frac{\Phi_u \times \Phi_v(h(s,t))}{\|\Phi_u \times \Phi_v\| \cdot (h(s,t))} = -\frac{\tilde{\Phi}_s \times \tilde{\Phi}_t(s,t)}{\|\tilde{\Phi}_s \times \tilde{\Phi}_t\|(s,t)} \quad \forall (s,t) \in \tilde{D}$$

Lemma 33. Let $\tilde{\Phi}$ and Φ be as above with $h : D \to \tilde{D}$ such that $h(s,t) = (u(s,t), v(s.t)) \in C^1(\tilde{D}, D)$. Then h is orientation-preserving iff $\det(\frac{\partial(u,v)}{\partial(s,t)}) > 0$, and h is orientation-reversing iff $\det(\frac{\partial(u,v)}{\partial(s,t)}) < 0$.

Proposition 8. Let S be an oriented C^1 regular parameterized surface $S = \Phi(D)$ and $\vec{F} \circ \Phi \in C^0(D, \mathbb{R}^3)$. Then $\iint_S \vec{F} \cdot d\vec{S}$ is well defined independent of the choice of parameterization, modulo the choice of orientation, (+) or (-).

Corollary 7. $S = \Phi(D)$ an oriented regular parameterized surface. Then

$$\iint_{S^\mp} \vec{F} \cdot d\vec{S} = -\iint_{S^\pm} \vec{F} \cdot d\vec{S}$$

3.6. Green's Theorem

Definition 49 (Regular Surface)**.** $S \subset \mathbb{R}^3$ is a **regular surface** with C^1 boundary if $S \subset \mathbb{R}^3$ is compact and if $\forall p \in S$, $\exists r, \delta > 0$, $B_r(p) \subset \mathbb{R}^3$, and $D = \{-\delta < u < \delta, -\delta < v < \delta\} \subset \mathbb{R}^3$ and a regular map $\Phi : D \to \mathbb{R}^3$ such that either

(1) $S \cap B_r(p) = \Phi(D)$;

(2) $S \cap B_r(p) = \Phi(D^+)$, where $D^+ = \{-\delta < u < \delta, 0 \leq v < \delta\}$

Points in (1) are called **interior points** of S and points in (2) are called **boundary points** of S. $\partial S := \{\text{all boundary points of } S\}$

Example 3.5 (orientability)**.** Let $S^2_+ = \{x^2 + y^2 + z^2 = 1, z \geq 0\}$. We say that a regular surface S is **orientable** if there exists a unit normal $\vec{n}(p)$ at each $p \in S$ with $\vec{n} \in C^0(S, \mathbb{R}^3)$.

Definition 50. We say a bounded planar domain $D \subset \mathbb{R}^2$ is $(+)$ **positively oriented** if ∂D is oriented counter-clockwise, i.e. if the unit normal is $\vec{k} = (0, 0, 1)$.

Theorem 34 (Green's Theorem)**.** Let $D \subseteq \mathbb{R}^2$ be a domain with piecewise C^1 boundary ∂D, and $\vec{F} = (P, Q) : D \to \mathbb{R}^2$ a C^1 vector field on D. Then

$$\int_{\partial D} \vec{F} \cdot \vec{ds} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$$

Corollary 8.

$$\int_{\partial D} x \, dy = -\int_{\partial D} y \, dx = \frac{1}{2} \int_{\partial D} x \, dy - y \, dx = \text{area}(D)$$

Example 3.6. Compute the area of the ellipse $\left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}$ using Green.

To solve, notice the boundary ∂D is the ellipse $\left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right\}$. We can parameterize ∂D by $\gamma(t) = (a \cos t, b \sin t)$, $t \in [0, 2\pi]$. Then

$$\int_{\partial D} x \, dy = \int_0^{2\pi} a \cos t \cdot b \cos t \, dt = ab \int_0^{2\pi} \cos^2 t \, dt = ab\pi$$

Corollary 9. Let $D = D_1 \setminus D_2$ and ∂D_1, ∂D_2 piecewise C^1 with orientations induced from $D_1 + D_2$. Then if $\vec{F} = (P, Q)$ is a C^1 vector field on D with $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$ on D, then

$$\int_{\partial D_1} \vec{F} \cdot \vec{ds} = \int_{\partial D_2} \vec{F} \cdot \vec{ds}$$

Theorem 35. If $\vec{F} = (P, Q) \in C^1$, then

$$\int_{\partial D} \vec{F} \cdot \vec{n} \, ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) \, dA$$

3.7. Harmonic Functions

Definition 51 (Harmonic Function)**.** Let $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ be the Laplacian. $u \in C^2(D)$ is **harmonic** if $\Delta u = 0$ on D.

Corollary 10. u harmonic on domain $D \subset \mathbb{R}^2$ with piecewise C^1 boundary ∂D. Then

$$\iint_D \Delta u \, dA = \int_{\partial D} \Delta u \cdot \vec{n} \, ds = 0$$

Corollary 11 (Green's Identities)**.** Let $\varphi, \psi \in C^2(D)$, ∂D piecewise C^1 with positive orientation. Then

- $$\iint_D \varphi \Delta\psi \, dA = -\int_{\partial D} \nabla\varphi \cdot \nabla\psi \, dA + \iint_{\partial D} \varphi \nabla\psi \cdot \vec{n} \, dA$$
- $$\iint_D [\varphi\Delta\psi - \psi\Delta\varphi] \, dA = \int_{\partial D} (\varphi\nabla\psi - \psi\nabla\varphi \cdot \vec{n}) \, ds$$

3.8. Stokes' Theorem

Definition 52. Let $\vec{F} = (F_1, F_2, F_3) : \Omega \subset \mathbb{R}^3 \to \mathbb{R}^3$ be a C^1 vector field. Then the **curl** of $\vec{F}$ is defined as

$$\operatorname{curl} \vec{F} = \nabla \times \vec{F} = \det \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ F_1 & F_2 & F_3 \end{bmatrix} = \left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}, \frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}, \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} \right)$$

Theorem 36 (Stokes)**.** Let S be a compact regular oriented surface with $(+)$ orientation and piecewise C^1 boundary ∂S. Let $F : S \to \mathbb{R}^3$ be a C^1 vector field. Then

$$\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

Remark. Note,

$$\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = \iint_S \operatorname{curl} \vec{F} \cdot \vec{n} \, dS$$

Example 3.7. $\vec{F} = (P, Q, 0)$. Then $F_3 = 0$ and $P = P(x, y), Q = Q(x, y)$, so $\operatorname{curl} \vec{F} = \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$. Then we can see Green's theorem is a really just a special case of Stokes' theorem, since

$$\left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \cdot \vec{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Corollary 12. Suppose S_1, S_2 are two oriented surfaces with $\partial S_1 = \partial S_2 = \gamma$, and suppose S_1 and S_2 induce opposite orientations on γ. Then, for any $\vec{F} \in C^1$,

$$\iint_{S_1} \operatorname{curl} \vec{F} \cdot d\vec{S} = -\iint_{S_2} \operatorname{curl} \vec{F} \cdot d\vec{S}$$

Corollary 13. If S is a compact, oriented surface with $\partial S = \emptyset$ and $\vec{F} \in C^1(S)$, then

$$\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = 0$$

Corollary 14. Suppose $\vec{F} = \nabla F$ for some $f \in C^2$ and S is an oriented surface with $\partial S = \gamma_1 - \gamma_2$. Then,

$$\int_{\gamma_1} \vec{F} \cdot d\vec{s} = \int_{\gamma_2} \vec{F} \cdot d\vec{s}$$

Example 3.8. Let $C = \{y + z = 2\} \cap \{x^2 + y^2 = 1\}$ with $(+)$ orientation. Let $\vec{F} = (-y^2, x, z^2)$. Evaluate $\int_C \vec{F} \cdot d\vec{s}$.

To solve, by Stokes

$$\int_C \vec{F} \cdot d\vec{s} = \iint_S \operatorname{curl} \vec{F} \cdot d\vec{S}$$

Here $\operatorname{curl}\vec{F} = (0,0,1+2y)$ and $\vec{n} = \frac{(0,1,1)}{\sqrt{2}}$ Therefore,

$$\int_C \vec{F}\cdot d\vec{s} = \iint_D (0,0,1+2y)\cdot(0,1,1)\ dA = \iint_D (1+2y)\ dA$$

To solve this integral, we make the change of variables $x = r\cos\theta, y = r\sin\theta$, $D = \{0 \le r \le 1, 0 \le \theta \le 2\pi\}$. Then we get

$$= \int_0^{2\pi}\int_0^1 (1+2r\sin\theta)r\ dr\ d\theta = \int_0^{2\pi}\left(\frac{1}{2}+\frac{2}{3}\sin\theta\right)d\theta = \pi$$

3.9. Divergence Theorem

Theorem 37 (Divergence Theorem)**.** Let $W \subset \mathbb{R}^3$ be a bounded solid region in $\mathbb{R}^3$ with piecewise smooth boundary $S = \partial W$ with positive orientation and $\vec{F} \in C^1(W, \mathbb{R}^3)$. Then

$$\iiint_W \operatorname{div}\vec{F}\ dV = \iint_S \vec{F}\cdot d\vec{S}$$

Here, $\operatorname{div}\vec{F} := \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

Corollary 15. Under the same assumptions as above, for any $f, g \in C^2(W)$,

$$\iiint_W [g\Delta f - f\Delta g]\ dV = \iint_{\partial W}\left[g\nabla f\cdot d\vec{S} - f\nabla g\cdot d\vec{S}\right]$$

Corollary 16. Suppose $W, W_1, W_2, \ldots, W_k \subset \mathbb{R}^n$ are regions as in the divergence theorem with

- $W_i \cap W_j = \emptyset$ for all $i \ne j$;
- $W_j \subset W^{\text{int}}$, $p_j \in W_j^{\text{int}}$ for all $j = 1, \ldots, k$;

for some points $p_1, \ldots, p_k$. If $\vec{F} \in C^1(W \setminus \{p_1, \ldots, p_k\})$ and $\operatorname{div}\vec{F} = 0$ for all $x \in W \setminus \{p_1, \ldots, p_k\}$, then

$$\iint_{\partial W} \vec{F}\cdot d\vec{S} = \sum_{j=1}^k \iint_{\partial W_j} \vec{F}\cdot d\vec{S}$$

Example 3.9 (Gauss law)**.** Suppose $W \subset \mathbb{R}^3$ as in the divergence theorem and $\vec{F} = (x,y,z)$ and $r = \sqrt{x^2+y^2+z^2}$. Then, if $(0,0,0) \notin \partial W$,

$$\iint_{\partial W} \frac{\vec{F}}{r^3}\cdot d\vec{S} = \begin{cases} 4\pi & \text{if } (0,0,0) \in W^{\text{int}} \\ 0 & \text{otherwise} \end{cases}$$

To show this, set $\vec{F} = \frac{\vec{r}}{r^3}$. Then $\operatorname{div}\vec{F} = 0$, $\vec{r} \in \mathbb{R}^3 \setminus (0,0,0)$. SO if $(0,0,0) \notin W$, $\vec{F} \in C^1(W)$ and by the divergence theorem, $\iint_{\partial W} \frac{\vec{r}}{r^3}\cdot d\vec{S} = 0$. If $(0,0,0) \in W$, pick $\delta > 0$ small so that $B_\delta(0) \subset W^{\text{int}}$. Let $\tilde{W} = W \setminus B_\delta(0)$. By the most recent corollary,

$$\iint_{\partial W} \frac{\vec{r}}{r^3}\cdot d\vec{S} = \iint_{\partial B_\delta(0)} \frac{\vec{r}}{r^3}\cdot d\vec{S} = \iint_{\partial B_\delta(0)} \frac{\vec{r}}{\delta^3}\cdot d\vec{S} = \frac{1}{\delta^3}\iint_{\partial B_\delta(0)} \vec{r}\cdot d\vec{S} = 4\pi$$

3.10. Divergence Theorem Applications

Example 3.10. Compute $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = (3x + z^{77}, y^2 - sinx^2z, xz + ye^{x^3})$, $S = \partial W$ where $W = \{0 \leq x \leq 1, 0 \leq y \leq 3, 0 \leq z \leq 2\}$.

To solve, we can use the divergence theorem. We have $\operatorname{div} F = 3 + 2y + x$. Then by the divergence theorem,

$$\begin{aligned}
\iint_S \vec{F} \cdot d\vec{S} = \iiint_W (3 + 2y + x)\, dV &= \int_0^1 \int_0^3 \int_0^2 (3 + 2y + x)\, dz\, dy\, dx \\
&= \int_0^1 \int_0^3 2(3 + 2y + x)\, dy\, dx \\
&= \int_0^1 2(9 + 9 + 3x)\, dx \\
&= \int_0^1 36 + 6x\, dx = 39
\end{aligned}$$